

4. PROJECT DESIGN

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Team ID	SWTID-2026-4302
Project Name	Import Data using Transform Maps (Spreadsheet)
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4.1 Problem Solution Fit

Problem-Solution Fit means the team has found a real problem with its customer and the designed solution actually solves it. The canvas below summarises the fit (template: Problem-Solution Fit canvas, Daria Nepriakhina / Amaltama.com, CC BY-NC-ND 4.0).

Purpose / Vision: Load employee data from spreadsheets into ServiceNow quickly, accurately and without duplicates, and make it visible through reports.

1. Customer Segment(s) [CS]	6. Customer Constraints [CC]	5. Available Solutions [AS]
ServiceNow administrators and HR data owners who receive employee data in Excel / Google Sheets.	Headers must match target fields; administrator access and Import Sets are required; limited time; PDI environment.	Manual entry form by form (slow, error-prone); custom scripts / REST integrations (need developer skill); importing without coalesce (creates duplicates).
2. Jobs-to-be-done / Problems [J&P]	9. Problem Root Cause [RC]	7. Behaviour [BE]
Import bulk employee data accurately; update records when corrected data arrives; avoid duplicate employees; see summaries by department and location.	Employee data originates outside ServiceNow in spreadsheets and is re-sent with changes; with no unique-key matching, every import inserts new rows.	Keep data in Excel, re-upload corrected sheets, open the Employee table to check results manually.
3. Triggers [TR]	10. Your Solution [SL]	8. Channels of Behaviour [CH]
A new batch of employee data arrives; a corrected sheet is received; HR asks for headcount by department or location.	Load Data into an Import Set table (Employee Import) + Transform Map (Sample Spreadsheet Import) with Coalesce on Employee ID into Employee Test; Transform History to verify; Reports and Dashboard for insight.	8.1 Online: Load Data, Transform Maps, Transform History, Reports and Dashboards in ServiceNow. 8.2 Offline: Spreadsheets shared by email or HR teams; conversations to confirm corrections.
4. Emotions: Before / After [EM]		
Before: frustrated, worried about mistakes and duplicates. After: confident, in control, able to trust the employee data.		

4.2 Proposed Solution

S.N o.	Parameter	Description
1	Problem Statement (Problem to be solved)	Bulk employee data (Employee ID, Name, Email, Department, Location) arrives in spreadsheet format and must be migrated into ServiceNow efficiently and accurately. Manual entry is slow and error-prone, and repeated imports of the same data can create duplicate records.
2	Idea / Solution description	Upload the spreadsheet with Load Data into a staging Import Set table (Employee Import, u_employee_import). A Transform Map (Sample Spreadsheet Import) maps the fields to the Employee Test table (u_employee_test). Coalesce on Employee ID updates existing records and inserts new ones. Results are checked in Transform History and summarised in three reports on the Employee Analytics Dashboards.
3	Novelty / Uniqueness	Combines the full import cycle in one small project: load, map, transform, validate, coalesce-based duplicate prevention and reporting. Re-importing the same sheet changes nothing (0 inserted, 0 updated, 4 ignored in the test run), so imports can safely be repeated.
4	Social Impact / Customer Satisfaction	Saves administrators and HR staff from repetitive data entry, reduces errors and gives trustworthy employee data and clear visibility for decision making.
5	Business Model (Revenue Model)	Internal efficiency solution rather than a sold product: value comes from time saved and improved data quality. The same pattern can be offered as a reusable import accelerator by an implementation team.
6	Scalability of the Solution	The staging table + transform map pattern works for larger files and other target tables (for example users or configuration items), and imports can later be automated with data sources and schedules.

4.3 Solution Architecture

Solution architecture bridges the gap between the business problem and the technology solution: it describes the structure and behaviour of the solution, its features and its requirements. The architecture below shows the flow of employee data from Google Sheets through ServiceNow to HR users.

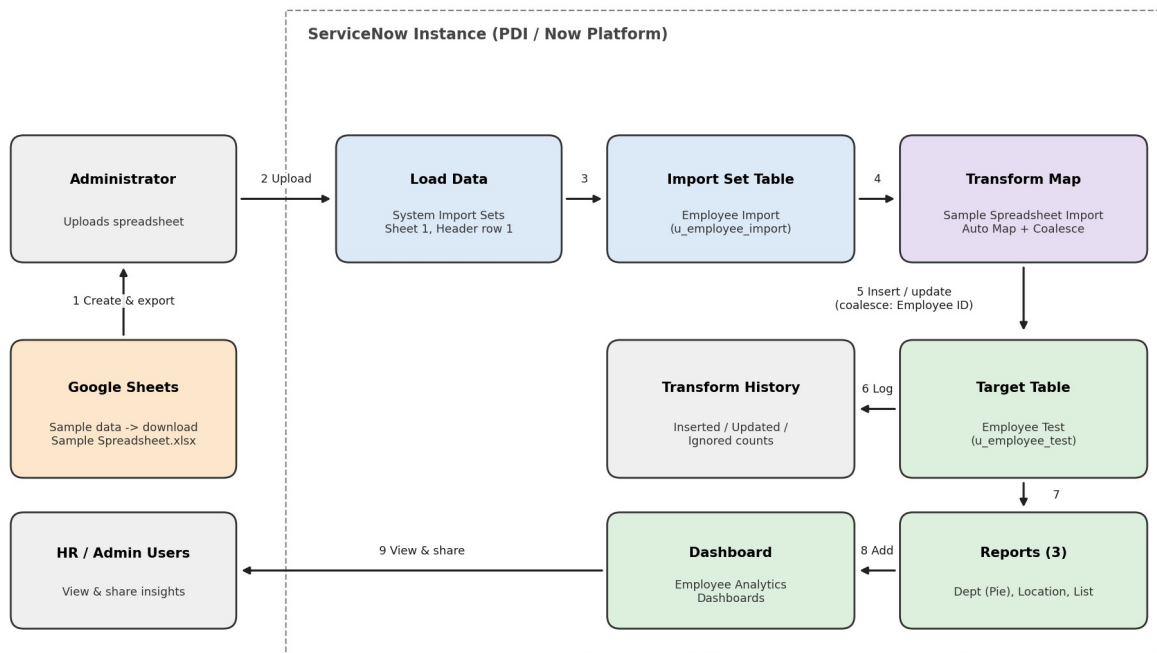


Figure 4.1 – Architecture and data flow of the employee data import solution on ServiceNow

Data flow (numbers match the figure)

Step	Description
1	The administrator creates sample employee data in Google Sheets and downloads it as Sample Spreadsheet.xlsx.
2	The file is uploaded with Load Data (System Import Sets): Sheet number 1, Header row 1.
3	Rows are stored in the Employee Import staging table (u_employee_import).
4	The Transform Map Sample Spreadsheet Import maps source fields to Employee Test fields (Auto Map Matching Fields).
5	On Transform, records are inserted or updated in Employee Test (u_employee_test); Coalesce on Employee ID prevents duplicates.
6	Each run is logged in Transform History with inserted / updated / ignored counts.
7–8	Three reports are built on Employee Test and added to Employee Analytics Dashboards.
9	HR and administrators view and share the dashboard.

Key configuration objects

Object	Name	Notes
Target table	Employee Test (u_employee_test)	Five String fields: Employee ID, Employee Name, Email, Department, Location
Import set (staging) table	Employee Import (u_employee_import)	Created from Load Data; temporary holding area
Transform map	Sample Spreadsheet Import	Source: Employee Import; Target: Employee Test; Auto Map Matching Fields
Coalesce field	Employee ID	Coalesce = true on the Employee ID field map

Object	Name	Notes
Reports	Employees by Department (pie, Count); Employees by Location; Employee List	Built on Employee Test
Dashboard	Employee Analytics Dashboards	Created via PA_DASHBOARDS.FORM after enabling Admin Overrides on the create ACL